

System Driven Hardware Design

-Doppler Shift-

International Master of Science

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This is the physical principal of our technical system. Hence it should be well understood.

Agenda

- o Christian Doppler
- o Derivation of Doppler Shift
- o Expected Doppler Shift by Movements of objects

Where did you get in touch with the Doppler shift?

Brainstorm

Where do you observe Doppler Shifts?

In which case do you have an increase of frequency?

In which case do you have a decrease of frequency?

Do you know technical applications of the Doppler shift?

[Sound Example](#)



Christian Andreas Doppler (* 29. November 1803 in Salzburg^[1]; † 17. März 1853 in Venedig)

Doppler realized, if there is a relative motion between two stars, there is a change in the colour observed on the earth.

Finally his Hypothesis was experimentally proven, due to the invention of the steam train:

*If a sound source and receiver **move towards** each other, you observe an **increase** in the sound frequency.*

*If a sound source and receiver **depart** from each other, you observe an **decrease** in the sound frequency.*

Higher frequency light waves means blue color
Lower frequency light waves means red color in visible light spectrum.
This is what known as a Redshift phenomenon

Assumption: The waves propagate in a non moving media with a fixed velocity of propagation.

In general

Fixed Source

$$f = f_o \left(1 \pm \frac{v}{c}\right)$$

- + moving towards
- departing

Moved Source

$$f = f_o \frac{1}{1 \mp \frac{v}{c}}$$

- moving towards
- + departing

- f_o : source frequency
- f : observed frequency of the receiver
- v : relative velocity
- c : wave propagation velocity

If $v \ll c$ holds

$$f = f_o \left(1 \pm \frac{v}{c}\right)$$

Literature

- “*Physik für Studenten der Natur- und Technikwissenschaften*”, Fachbuchverlag Leipzig, Heribert Stroppe

Youtube Video:

- “[The Doppler Effect_ what does motion do to waves](https://www.youtube.com/watch?v=...)“, by <https://www.patreon.com/AltShiftX>